

Veera Ajay Vardhan Medisetty

+1 (409) 665-0730 | ajayvardhanmedisetty@gmail.com | Beaumont, TX.

Summary:

Experienced industrial engineering graduate with 1.8 years of experience in Manufacturing Engineering, Design, and Quality Engineering. Skilled in optimizing manufacturing processes, designing efficient production layouts, and ensuring high quality standards. Seeking a challenging role where I can contribute my knowledge of mechanical and industrial engineering to drive operational excellence and continuous improvement in a dynamic work environment.

Skills:

Software & Design: ANSYS, Fusion 360, SAP, Solid Works.

Technical Documentations: Standard Operating Procedures (SOP), Work Instructions, Reports & Process Flowcharts.

Safety & Compliance: Safety Regulations, Safety Audits, Compliance with Industry Standards.

Education:

Master of Science in Industrial Engineering.

Mar - 2024

LAMAR University, Texas, United State

Bachelor of Technology in Mechanical Engineering

Aug - 2020

NRI Institute of Technology, Andhra Pradesh, India.

Board of Intermediate

Apr - 2016

Sri Chaitanya, Vijayawada, India.

Experience:

Manufacturing Engineer

Jun 2021 – Sep 2022

Hyderabad, India.

- Reduced product defects by 20% through the implementation of SPC techniques and thorough root cause analyses.
- Improved workflow efficiency by 15% and decreased process cycle time by 12% using Lean principles such as 5S & Kaizen.
- Addressed complex manufacturing issues, cutting machine downtime by 20% and increasing production output by 10%.
- Created and maintained SOPs and work instructions, enhancing process consistency and aiding in new employee training.
- Ensured adherence to safety regulations and conducted regular safety audits, leading to a safer work environment and a 25% reduction in safety incidents.

Projects:

3D Printing Project

- Led a project to develop and prototype custom components using 3D printing technology, aimed at enhancing manufacturing efficiency and reducing costs.
- Managed the 3D printing workflow, including material selection, printer calibration, and post processing to ensure high quality outputs.
- Achieved 25% reduction in production costs by optimizing material usage and streamlining the printing process.
- Reduced prototype development time by 40% through effective project management and implementation of 3D printing solutions.
- Coordinated with cross functional teams to integrate 3D-printed components into existing systems, resulting

2 Stroke & 4 Stroke IC Engines

- The project focused on the comparative performance evolution of both engine types, highlighting their applications in various mechanical systems.
- Conducted in-depth analysis of the working principles, design and efficiency of 2-stroke & 4 stroke combustion engines.
- Evaluated the fuel efficiency, power output, emissions of both engine types under various operating conditions.
- Prepared detail reports and presentations summarizing the findings and recommendations for industrial applications.

CFD-CFX Analysis Of A Single - Phase Mixing Of Fluids Without Aid Of Stirrers

- The project aimed at optimizing fluid mixing processes in industrial applications where mechanical stirring is impractical or inefficient.
- Performed Computational Fluid Dynamics (CFD) Analysis using ANSYS CFX to study the mixing behavior of single-phase fluids in the absence of stirrers.
- Analyzed the flow patterns, turbulence, and velocity profiles of fluids in various container geometries.
- Identified key parameters influencing mixing efficiency, such as fluid viscosity, container shape, and inlet velocity.
- Focused on applications where mechanical stirring is impractical, such as chemical reactors, bioreactors, and food processing.
- Utilized statistical methods and statistical quality control techniques (7 Quality tools).